

CYS401: Fundamentals of Cybersecurity

Class Activity 5

Objectives

Research and analyze cyber security incidents.

Part 1: Conduct search of high profile cyberattacks.

Part 2: Write an analysis of a cyberattack.

Background / Scenario

Governments, businesses, and individual users are increasingly the targets of cyberattacks and experts predict that these attacks are likely to increase in the future. Cybersecurity education is a top international priority as high-profile cyber-security related incidents raise the fear that attacks could threaten the global economy. The Center for Strategic and International Studies estimates that the cost of cybercrime to the global economy is more than \$600 billion annually. In this lab, you will study four high profile cyberattacks and be prepared to discuss the who, what, why and how of each attack.

Instructions

Part 1: Conduct search of high profile cyberattacks.

- a. Conduct a search for each of the cyberattacks listed below. Your search will likely turn up multiple results ranging from news articles to technical articles.
 - The Stuxnet Virus
 - Microsoft customer support database breach
 - Marriott data breach
 - Lifelabs data breach
 - United Nations data breach
- b. Read the articles found from your search in Step 1a and be prepared to discuss and share your research on the who, what, when, where, and why of each attack.

Part 2: Write an analysis of a cyberattack.

Select one of the high-profile cyberattacks from Step 1a and write an analysis of the attack that includes answers to the questions below.

Questions:

- a. Who were the victims of the attacks?

The primary victims were Iranian nuclear facilities, particularly the Natanz uranium enrichment facility. The attack targeted industrial systems responsible for operating uranium enrichment centrifuges.

- b. What technologies and tools were used in the attack?

The attackers used the **Stuxnet computer worm**, which exploited multiple Windows zero-day vulnerabilities. The malware spread through infected USB drives and compromised computers. It specifically targeted Siemens Step7 software and programmable logic controllers (PLCs) used to control industrial equipment.

- c. When did the attack happen within the network?

Stuxnet was discovered in June 2010, although infections had occurred earlier. The malware entered Windows systems, spread through connected devices and networks, and eventually reached the industrial control systems.

- d. What systems were targeted?

The attack targeted **Siemens industrial control systems**, particularly programmable logic controllers (PLCs) responsible for controlling uranium enrichment centrifuges at Iran's Natanz nuclear facility.

- e. What was the motivation of the attackers in this case? What did they hope to achieve?

The attackers aimed to **sabotage Iran's nuclear enrichment program** by disrupting the operation of uranium enrichment centrifuges.

The objective was to damage equipment and delay Iran's nuclear activities without conducting a conventional military attack.

f. What was the outcome of the attack? (stolen data, ransom, system damage, etc.)

Stuxnet caused physical damage to approximately **1,000 uranium enrichment centrifuges**, disrupting operations at the Natanz facility.

The attack demonstrated that malware could cause physical destruction by manipulating industrial control systems, making it one of the most significant examples of cyber sabotage.

References

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